

I. Introduction and Global Attention

- **Security Threat:** Climate change and environmental degradation are increasingly recognized as major threats to national and international security, not just ecological issues.
- **Global Prioritization:**
 - About 70 UN member states prioritized climate security risks at the 2015 Paris Conference.
 - The UN Secretary-General has urged the Security Council to allocate resources for mitigating these security threats.
- **Institutional Response:** The Intergovernmental Panel on Climate Change (IPCC) was established by the UN Environment Program and World Meteorological Organization to provide up-to-date data and guide policy formulation.

II. Definitions

- **Climate Change:**
 - **General:** Alteration of the environment due to human activities (waste disposal, fuel combustion, deforestation) that increase Greenhouse Gases (GHG).
 - **UNFCCC Definition:** Change in global atmosphere composition attributed directly or indirectly to human activity, superimposed on natural climate variability.
- **Environmental Degradation:**
 - The degeneration of the ecosystem due to the depletion of natural resources (water, soil, air).
 - A decline in the environment's capacity to support social and ecological needs.

III. Current Scenario and Statistics

- **Rising Temperatures:**
 - 11 of the 12 years between 1995–2006 were the warmest since 1885.
 - Temperatures are estimated to rise by **1.1–6.4°C** by the year 2100.
- **Disasters & Economic Loss:**

- 3,852 natural catastrophes have killed over 780,000 people and affected 2 billion others.
- Economic property losses exceed **\$960 billion**.
- Disasters are predicted to double in the next 10–15 years.
- **Glacial and Sea Level Changes:**
 - Sea levels have risen by **1.7mm/year** over the past century due to melting land ice and mountain glaciers.
 - Snow in the northern hemisphere has sharply retreated, especially in spring and summer.

IV. Causes of Climate Change and Degradation

- **Greenhouse Gases (GHG):**
 - The accumulation of GHGs (Carbon Dioxide, Methane, Nitrous Oxide) forms a "blanket" that traps solar radiation (Global Warming).
 - **CO2:** The largest contributor; its accumulation has increased by 70%, with over half occurring since 1970 due to energy consumption.
- **Human Activities:** The primary drivers of change include:
 - **Combustion of Fossil Fuels:** Releases carbon compounds.
 - **Deforestation:** Destroys natural carbon sinks (trees), increasing atmospheric CO2 and leading to soil erosion.
 - **Urbanization & Overpopulation:** Creates demand for resources, leading to over-exploitation and destruction of forests.
 - **Industrialization:** Releases toxic gases and accelerates GHG accumulation.

V. Climate Variability as a Risk to Global Peace

The document defines security not just as military defense, but as protection from hunger, disease, and environmental hazards.

A. Migration and Displacement

- **Refugees:** In 2001, 25 million people were refugees due to climatic catastrophes. This is predicted to rise to **50 million by 2050**.

- **Mass Migration:** Driven by droughts, storms, and sea-level rise, forcing people to move to safer regions. Between 1979-2008, 1.6 billion people suffered from droughts.

B. Resource Conflicts (Water & Land)

- **Water Scarcity:**
 - Over **1.1 billion** people lack access to safe water; this causes 3.5 million deaths annually (mostly children).
 - By 2020, 120 million to 1.2 billion people in South/South East Asia will face water scarcity.
 - 300 potential water conflicts exist globally (e.g., Indo-Pak water conflict).
- **Herders vs. Farmers (Nigeria):** Climate change drives herders to encroach on farmland, causing violent conflicts that kill thousands (approx. 2,000/year between 2011-2016), exceeding Boko Haram's toll in some years.

C. Food Insecurity

- **Hunger:** If global warming increases by 3°C, the number of people stricken by hunger could rise from 250 million to **550 million**.
- **Production Drop:** By 2020, 50% of rain-fed arable land in some regions may be lost. By 2050, global rice production may drop by 10% and wheat by 30%.

D. Infrastructure and Economic Instability

- **Facility Destruction:** Climate impacts destroy energy plants. Example: The Japan earthquake caused the Fukushima nuclear explosion, disrupting public order.
- **Adaptation Costs:** Developing countries need **\$50 billion** to prepare for climatic catastrophes.

VI. Specific Country Case Studies

- **Bangladesh:**
 - Most predisposed to climate risks.
 - **Arsenic Crisis:** Floods contaminate water, exposing 77 million people to arsenic, potentially causing 200,000+ deaths.
 - **Displacement:** A projected 2-3 ft sea-level rise could displace **30 million** people.

- **Sudan (Darfur):** The war was partly caused by drought and land deterioration, which eroded topsoil and forced migration from North to South, triggering conflict.
- **Haiti:** In 2008, violent protests erupted due to environmental destruction. Forest cover dropped from 60% (1923) to **1.5% (2008)**.

VII. Conclusion and Recommendations

- **Threat Assessment:** Climate change threatens international peace, increases budget expenditures, and endangers health and livelihoods.
- **Vulnerability:** Developing nations are most vulnerable due to a lack of capacity to handle calamities.
- **Action Required:**
 - **Alliance:** Countries must define individual and collective roles to reduce threats.
 - **Accountability:** International agencies should hold countries responsible. **Trading restrictions** or sanctions could be imposed on "defiant nations" that fail to act.
 - **Penalties:** National agencies should penalize those who violate environmental guidelines.